

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20		VB2					***** BALL VALVES *****			B1	ALL BOLTS ARE METRIC DIA.
30										B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1
30	U04		MODULAR DBB	M03AC02	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0		INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
40										FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
40	U04		BALL	A03AC01	3	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0		USE 1 1/2IN WITH THERMOWELL
50	U04		BALL	A03AC02	3	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO	0	0	UJ2	CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH
60	U04		BALL	A03AC07	6	6	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0		CHEMICAL SEALS.
70	U04		BALL	A03AC08	6	6	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0		USE THESE VALVES ONLY WHERE SPECIFIED
80	U04	VA5	BALL	B03AC01	0.75	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		ON PIPING AND INSTRUMENT DIAGRAMS
90	U04	VA5	BALL	B03AC02	0.75	3	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0	VA5	SOFT SEATED BALL VALVES ARE LIMITED TO
100	U04	VA5	BALL	B03AC03	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB GO	0	0		MAXIMUM TEMPERATURE OF 200°C.
110	U04	VA5	BALL	B03AC04	4	4	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB GO	0	0	VB2	SPRING PISTON CHECK VALVES ARE PREFERED
120	U04	VA5	BALL	B03AC07	6	6	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		IN THE HORIZONTAL POSITION, BUT MAY BE
130	U04	VA5	BALL	B03AC08	6	6	BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0	VC1	USED IN THE VERTICAL POSITION WITH
140							***** THROUGH CONDUIT GATE VALVES *****				ENGINEERING APPROVAL.
140											
150			GATE	W03AC70	3	6	GATE 300# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
160							***** GLOBE VALVES *****				
160											
170											
170	U04		GLOBE	L03AC48	0.75	1.5	GLOBE 300# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW	0	0		
180			GLOBE	L03AC41	2	6	GLOBE 300# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
190							***** CHECK VALVES *****				
190											
200											
200	U04	VC1	CHECK	C03AC50	0.75	1.5	CHECK PISTON 300# RF LTCS A350 LF2 CL.1 TRIM 16	0	0		
210			CHECK	C03AC45	2	6	CHECK DUAL WAFER LUG 300# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0		
220							***** PIPE *****				
220											
230											
230			PIPE	5358512	0.75	0.75	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0		
240			PIPE	5358508	1	1.5	PIPE SCH XS SMLS LTCS A333 GR.6 PE	0	0		
250			PIPE	5358514	1	2	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
260			PIPE	5358505	3	6	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0		
270		812	PIPE	5358513	0.5	0.5	PIPE SCH XXS SMLS LTCS A333 GR.6 BE	0	0		
280							***** FITTINGS *****				
280											
290											
290			ELBOW	5392816	0.75	0.75	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0		
300			ELBOW	5392600	1	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0		
310			ELBOW	5392438	3	6	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0		
320			ELBOW	5392822	0.75	0.75	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6	0	0		
330			ELBOW	5392606	1	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0		
340			ELBOW	5392444	3	6	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0		
350											
350			TEE	5392825	0.75	0.75	TEE BW SCH 160 LTCS A420 WPL6	0	0		
360			TEE	5392609	1	2	TEE BW SCH XS LTCS A420 WPL6	0	0		
370			TEE	5392447	3	6	TEE BW SCH STD LTCS A420 WPL6	0	0		
380			TEE RED	5392469	0.75	0.75	TEE RED BW SCH XS/160 LTCS A420 WPL6	1	2		
390			TEE RED	5392661	1	1.5	TEE RED BW SCH XS LTCS A420 WPL6	1.5	2		
400			TEE RED	5392454	1.5	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	4		
410			TEE RED	5392445	3	4	TEE RED BW SCH STD LTCS A420 WPL6	4	6		
420											
420			CAP	5392826	0.75	0.75	CAP BW SCH 160 LTCS A420 WPL6	0	0		
430			CAP	5392610	1	2	CAP BW SCH XS LTCS A420 WPL6	0	0		
440			CAP	5392448	3	6	CAP BW SCH STD LTCS A420 WPL6	0	0		
450											
450			RED CONC	5392579	0.75	0.75	REDUCER CONC SCH XS/160 LTCS A420 WPL6	1	2		
460			RED CONC	5392458	1	1.5	REDUCER CONC SCH XS LTCS A420 WPL6	1.5	2		
470			RED CONC	5392452	1.5	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	4		
480			RED CONC	5392449	3	4	REDUCER CONC SCH STD LTCS A420 WPL6	4	6		
490			RED ECC	5392465	0.75	0.75	REDUCER ECC SCH XS/160 LTCS A420 WPL6	1	2		
500			RED ECC	5392461	1	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	1.5	2		
510			RED ECC	5392453	1.5	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	4		
520			RED ECC	5392450	3	4	REDUCER ECC SCH STD LTCS A420 WPL6	4	6		
530											
530	U04		OLETS	5504450	0.75	0.75	NIPOFLANGE 300# RF SCH XS/160 LTCS A350 LF2 CL.1	2	2		
540	U04		OLETS	5504451	0.75	0.75	NIPOFLANGE 300# RF SCH STD/SCH 160 LTCS A350 LF2 CL.1	3	6		
550	U04		OLETS	5504020	1	1.5	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2 CL.1	2	2		
560	U04		OLETS	5504021	1	2	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2 CL.1	3	6		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - REQUIRED PIPE BENDS - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 343 °C COMMENT: REFER TO NOTE 800	12/16/2024	BY OZAC	U04	Page 1 of 2	300K5D

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570	U04	UJ2	OLETS	5504025	1.5	1.5	NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	4	6	
580										
580	U04		OLETS	5505192	0.75	0.75	WELDOLET STD/160 LTCS A350 LF2 CL.1	3	6	
590	U04		OLETS	5505179	1	2	WELDOLET SCH STD/XS LTCS A350 LF2 CL.1	3	6	
600										
600	U04	UJ2	ELBOFLG	5516625	1.5	1.5	ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1	4	6	
610										
610							***** FLANGES *****			
620										
620	U04	FA8	TEE RED	5599703	0.5	0.5	FLG SCRDR (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1	0.75	1.5	
630	U04		FLANGE	5556480	0.5	0.75	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 160	0	0	
640	U04		FLANGE	5556476	1	2	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
650	U04		FLANGE	5556473	3	6	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
660	U04		BLIND	5615576	0.75	6	FLG BLIND 300# RF LTCS A350 LF2 CL.1	0	0	
670										
670	U04		FLG ADAPTER	5851203	0.5	1.5	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0	
680										
680							***** ORIFICE FLANGES*****			
690										
690	U04		ORIFICE FLG	5623947	2	2	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
700	U04		ORIFICE FLG	5623944	3	6	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
710										
710							***** PIPE NIPPLES *****			
720										
720		812		0			BBE - TO BE CUT FROM PIPE			
730										
730							***** GASKETS *****			
740										
740			GASKET	5673033	0.5	6	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
750										
750		B1					***** BOLTING *****			
760										
760	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
770										
770							**** BLINDS & SPACERS ****			
780										
780			SPEC BLIND	5749581	0.75	6	SPECTACLE BLIND 300# RF LTCS A516 GR.70	0	0	
790										

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